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"""
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uma.semantic - UMA Semantic Engine (Product Engine).
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Architecture:

```
SemanticEngine → compiles intent into UMA_IR
UMA_IR          → execution graph
UMAExecutor     → drives UMAClient kernel
UMAClient       → physics (untouched by semantic logic)
```

Quick start::

```
from uma import Config, UMAClient
from uma.semantic import SemanticEngine

client = UMAClient(Config())
client.initialize(psi0)

engine = SemanticEngine(z_target=z_target)
result = engine.run(client, n_steps=200)
print(result.summary())
```

Text input via Engine3 bridge::

```
from uma.semantic import string_to_observation
obs, y = string_to_observation("your input text", dim=client.projection.real_
result = engine.run(client, n_steps=200, observations=[(obs, y)])
```

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"""
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```
from uma.semantic.friction import (
    SemanticFriction, SemanticFrictionConfig, FrictionRecord,
)
from uma.semantic.constraints import (
    EqualityConstraint, LinearConstraint, BallConstraint,
    QuadraticConstraint, ConstraintSet,
)
from uma.semantic.inarticulation import (
    ProductionMetrics, Inarticulator, NullInarticulator,
    ProjectionInarticulator, classify_stage,
)
from uma.semantic.ir import UMA_IR, IRNode
from uma.semantic.executor import UMAExecutor, ExecutorResult
from uma.semantic.engine import (
    SemanticEngine, SemanticEngineConfig, SemanticEngineResult,
    tokenize_to_binary_weight, string_to_observation,
```

)

```
__all__ = [  
    "SemanticFriction", "SemanticFrictionConfig", "FrictionRecord",  
    "EqualityConstraint", "LinearConstraint", "BallConstraint",  
    "QuadraticConstraint", "ConstraintSet",  
    "ProductionMetrics", "Inarticulator", "NullInarticulator",  
    "ProjectionInarticulator", "classify_stage",  
    "UMA_IR", "IRNode",  
    "UMAExecutor", "ExecutorResult",  
    "SemanticEngine", "SemanticEngineConfig", "SemanticEngineResult",  
    "tokenize_to_binary_weight", "string_to_observation",  
]
```